

CrohnIPI Diffusion: Class-Balanced Synthetic Augmentation for Crohn’s Endoscopy Classification

Sayan Nath ¹ Aditya Shekhar Nittala ¹

¹Department of Computer Science, University of Calgary, Calgary, Canada

Motivation

Clinical endoscopy datasets are often highly imbalanced, where majority classes dominate the training data. This imbalance makes it difficult for classification models to learn rare Crohn’s disease findings such as ulceration, edema, and stenosis.

Our work investigates whether diffusion-generated synthetic images can help improve classification performance under severe class imbalance.

Figure placeholder
dataset-fold-distribution

Figure 1. CrohnIPI dataset distribution across five folds. The normal class dominates the dataset, creating a strong class-imbalance problem.

Research Question

Key question: To what extent can per-class diffusion-generated synthetic images improve the classification performance of CrohnIPI models under severe class imbalance?

Dataset

Dataset: CrohnIPI endoscopy image dataset

Classes: N, E, O, AU, S, U3-10, U>10

Evaluation setup: 5-fold cross-validation

Main challenge: Severe class imbalance, where the majority class dominates and rare pathological classes have fewer examples.

Proposed Method

We train per-class LoRA diffusion adapters using real CrohnIPI images. Each adapter generates synthetic images for a specific clinical class. The generated images are then passed through multiple filtering steps before being used to train a ConvNeXt-Large classifier.

Figure placeholder
synthetic-augmentation-pipeline

Figure 2. Overview of the proposed synthetic augmentation pipeline using per-class LoRA diffusion adapters and filtered synthetic data for classifier training.

Synthetic Image Filtering

To reduce low-quality or unsafe synthetic samples, we apply several filtering steps before using generated images for downstream classifier training.

Filtering Step	Purpose
Artifact detection	Remove blurry or broken-color images
Class-consistency check	Remove images predicted as the wrong class
DINOv2 memorization check	Detect synthetic images that are too similar to real images
Diversity deduplication	Remove highly similar synthetic samples
Final filtered set	Used for downstream classifier training

Table 1. Synthetic image filtering steps used to remove low-quality, inconsistent, memorized, or duplicate generated samples.

Filtering thresholds:

- Laplacian variance < 50
- Channel std < 0.02
- ConvNeXt-Tiny confidence < 0.70
- Pairwise cosine similarity >= 0.95

Experiment Design

We compare real-only training against standard augmentation, oversampling, and multiple synthetic augmentation settings.

Condition	Description
Real only	Baseline using only real images
Real + Augmented	Standard clinical image augmentation
Oversampling	Repeated rare-class real images
Synthetic uncapped	Full per-class LoRA synthetic dataset
Synthetic balanced/capped	Equal synthetic cap per class
Focal loss	Synthetic data with focal loss
Inverse-frequency CE	Synthetic data with class-weighted loss

Table 2. Training conditions used to evaluate whether synthetic diffusion augmentation improves CrohnIPI classification under severe class imbalance.

Synthetic Dataset Design

Synthetic images were generated for each class using per-class LoRA adapters. We evaluate both uncapped synthetic datasets and balanced/capped synthetic datasets to study whether synthetic class balancing improves classification performance.

Figure placeholder
synthetic-images-per-class

Figure 3. Number of generated synthetic images per class across different synthetic dataset variants.

Main Results

Method	Macro F1	Balanced Accuracy	Macro AUROC
Real only	0.7888	0.7781	0.9709
Real + Augmented	0.7973	0.7902	0.9727
Oversampling	0.7786	0.7596	0.9628
Synthetic	0.8085	0.7949	0.9708
Synthetic + Balanced	0.8076	0.8090	0.9591
Focal Uncapped	0.8139	0.8078	0.9708
Focal Balanced	0.8077	0.7969	0.9721

Table 3. Classification performance across real-only, augmentation, oversampling, and synthetic augmentation conditions. Focal Uncapped achieves the best Macro F1.

Key result: Synthetic augmentation improves Macro F1 over the real-only baseline. The best Macro F1 is achieved using the Focal Uncapped synthetic condition.

Per-Class Recall Analysis

Synthetic augmentation improves recall for several rare classes, but the gains are not uniform across all labels. This suggests that diffusion-based augmentation can help with class imbalance, but class-specific generation quality remains important.

Figure placeholder
rare-class-recall

Figure 4. Rare-class recall across training conditions. Synthetic augmentation improves some rare-class recalls, but not all classes benefit equally.

Failure Analysis

The current method still struggles with the E class. Failure analysis shows that some E samples are confused with other classes, including the majority class. This indicates that synthetic data alone is not enough; better class-specific generation and expert-guided filtering may be needed.

Figure placeholder
e-class-failure-analysis

Figure 5. Failure analysis showing that the E class remains difficult under the current synthetic augmentation method.

Main Takeaways

- Per-class LoRA diffusion can improve CrohnIPI classification under severe class imbalance.
- Synthetic data performs better than simple oversampling.
- The best Macro F1 is achieved using synthetic data with focal loss.
- The E class remains challenging, showing the need for better class-specific generation and filtering.
- Future work should include stronger quality control and expert validation of synthetic endoscopy images.